PROJECT

#Chem LAB

Crafted By

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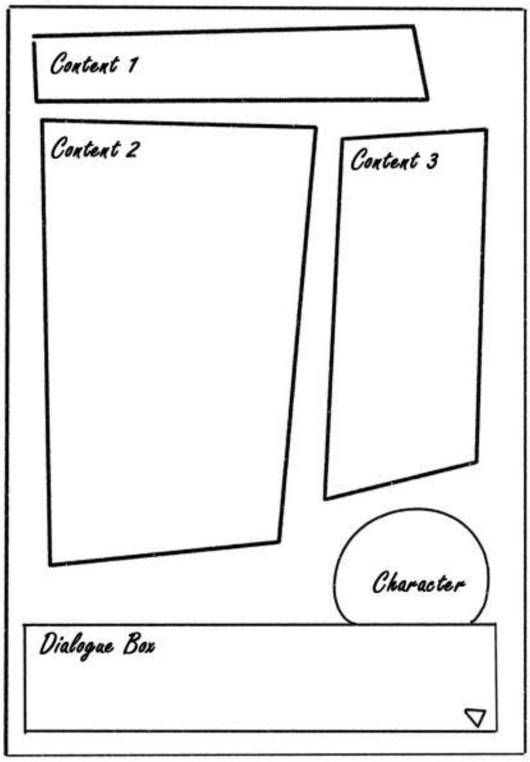
GAME OVERVIEW:

Title: Chem LAB
Genre: STEM
Platform: Mobile

Concept Overview:

Chem LAB is a fun and interactive chemistry **STEM** game where players mix chemicals, conduct experiments, and solve puzzles to learn about reactions, elements, and compounds.

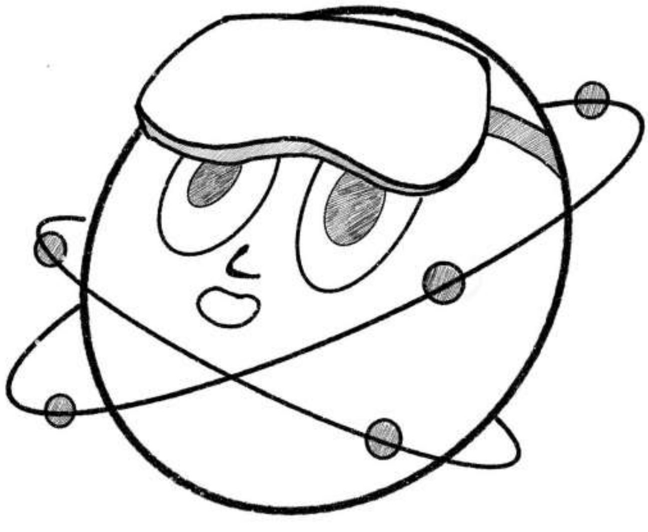
Layout of the game:



The game features a bright, playful layout set in a colorful cartoon lab where learning feels like play. The main screen has a cheerful scientist or robot guide, a circular reaction zone at the center where toy-like atoms float and combine, and fun animations when kids form correct molecules. The top bar shows the target molecule, score, and level, while the bottom has helpful icons and hint buttons. Each level feels lively with popping sounds, bouncing colors, and smiling characters. Between stages, short quiz and learning screens appear, letting kids tap or match cute toy versions of elements, making chemistry simple, friendly, and fun to explore.

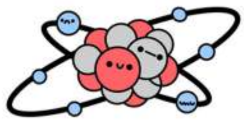
Character used in the game:

Lab Goggles



Atomic Character

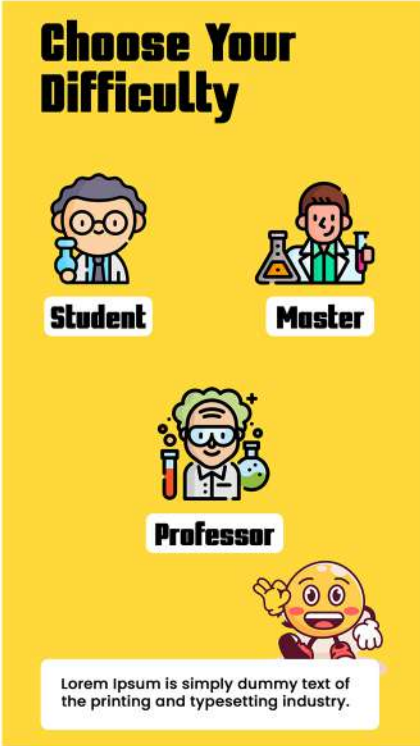
Reference Character



Atomic Structure

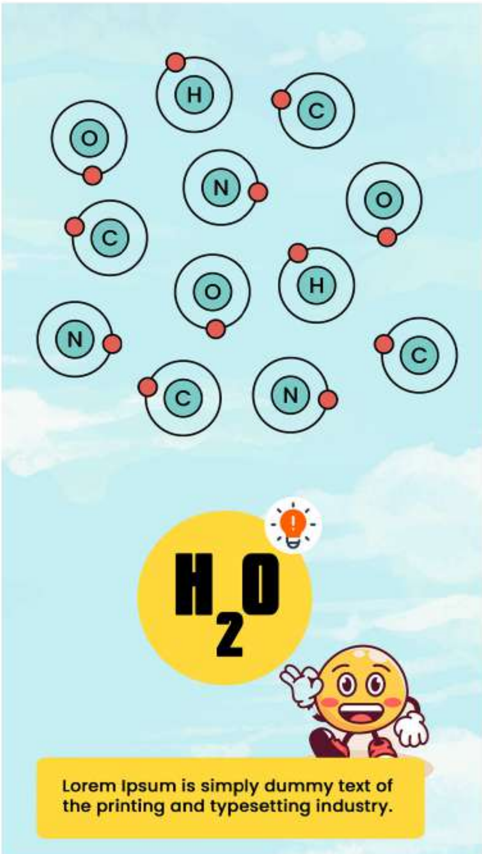
The atomic character is a cheerful, glowing little atom with playful spinning electrons and a bright smile. It acts as the player's friendly lab buddy, guiding and cheering them through each experiment, making chemistry feel fun, alive, and easy to learn.

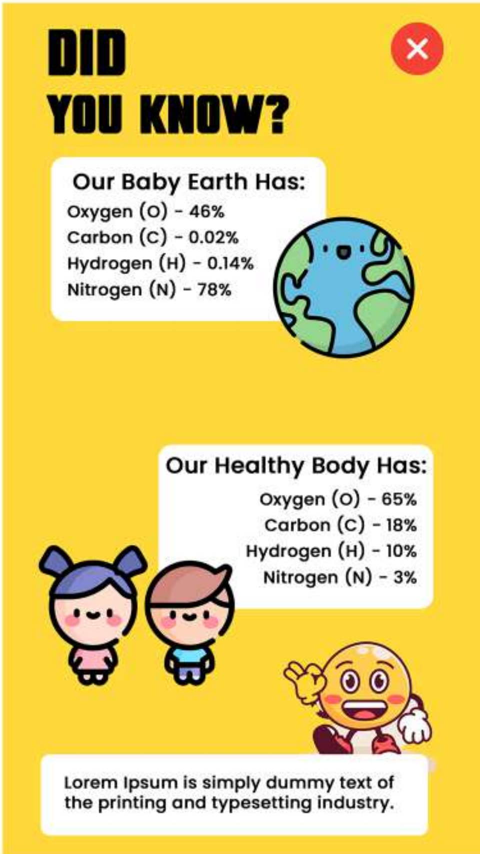
Design & Layout:



Introduction Page:

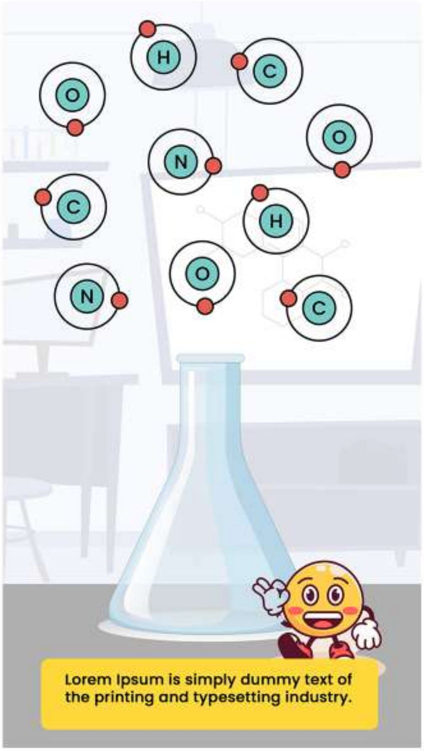
The new players are invited to the game with a simple designed home page with a play button. The very next nevel the players can choose the difficulty.

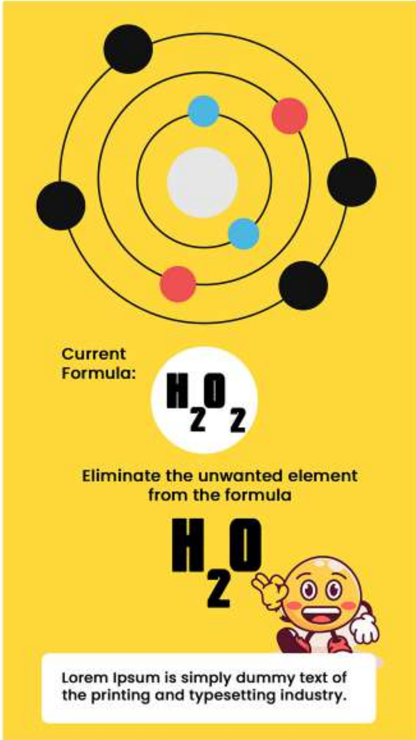




Level One:

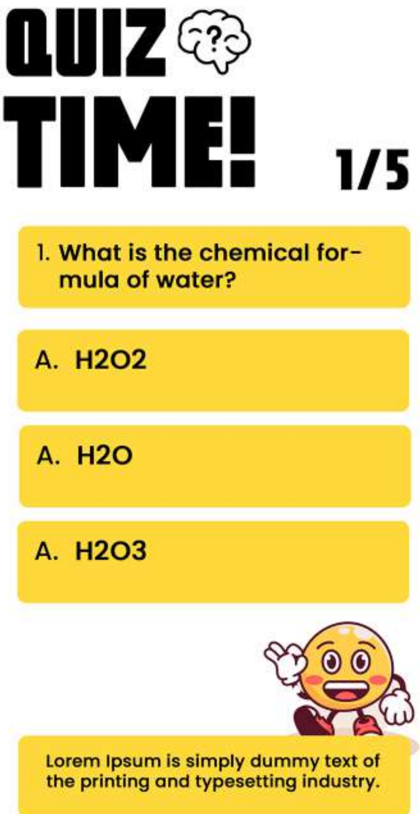
Here the players can select the atoms that are present on the screen re-alted to the chemical formula. The character in the bottom will give in-struction to you.You can also visit the **DID YOU KNOW** page just by pressing the bulb icon on the screen where it shares the importatnt and fun facts about the number of different types of atoms and molecules presented in both the earth and human body

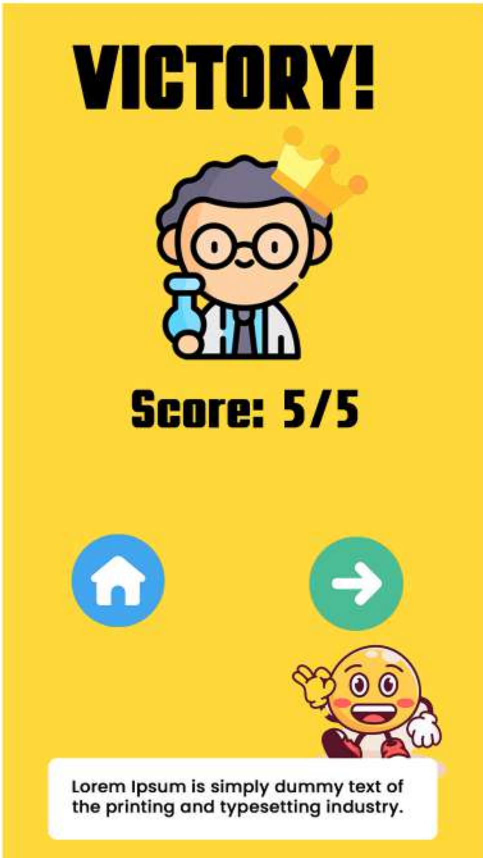




Level Two :

Now the players can go for the mixing stage where two types of atmos are added to the lab flask. Here the players can add botht the correct and incorrect amount of atoms so that they can understand the mistake of adding both unwanted and too much number of atoms. In the third level, the players can seen unwanted atoms, where the players can delete the unwated atoms just by pressing them. This way the players can easily learn the correct structure of the formula digitaly





Level Three:

Final stage will have a quiz section where players can have an opportunity toremember what they have learned for that day. Players can answer all the questions right so that they can land in the final page where they are appreaciatedfor being a part of the level. Now the players can visit the home page just byclicking the home icon

Color Theme - Yellow and White:

1. Importance of Yellow:

- **Symbol of Energy and Curiosity:** Yellow represents enthusiasm, creativity, and curiosity perfect for a STEM-based game that inspires learning.
- **Science Connection:** Often linked with lab lights, caution signs, and energy matching the chemistry lab atmosphere.
- **Attention-Grabbing:** Helps highlight important elements like buttons, alerts or experiment reactions.
- **Positive Mood:** Makes the game feel lively and welcoming, encouraging players to explore without fear of failure.

2. Importance of White:

- **Clean and Neutral Background:** Keeps the interface simple and clear, ensuring players focus on experiments and visuals.
- **Symbol of Purity and Clarity:** Reflects the idea of a clean laboratory environment and clear scientific understanding.
- **Balances Bright Colors:** White softens the intensity of yellow, creating a pleasant and readable contrast.
- **Professional Look:** Gives the game an educational, organized, and modern appearance.

3. Combined Usage:

- **Contrast and Readability:** Yellow elements (buttons, highlights) stand out well against white backgrounds.
- **Guided Focus:** Yellow draws the player's eyes to key actions or results, while white keeps the layout uncluttered.
- **Consistent Theme:** Together, they create a fresh, bright, and science-inspired design fitting for a chemistry learning game.
- **Emotional Impact:** The mix gives a sense of optimism, clarity, and safety ideal for young learners in a STEM environment

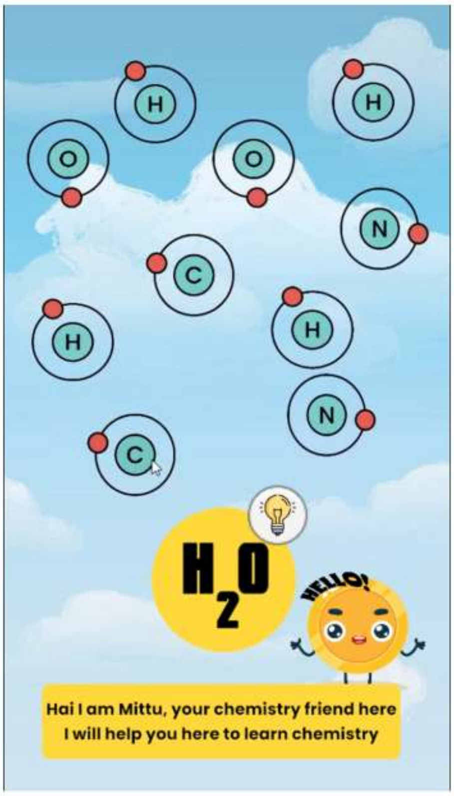
#ffd93b

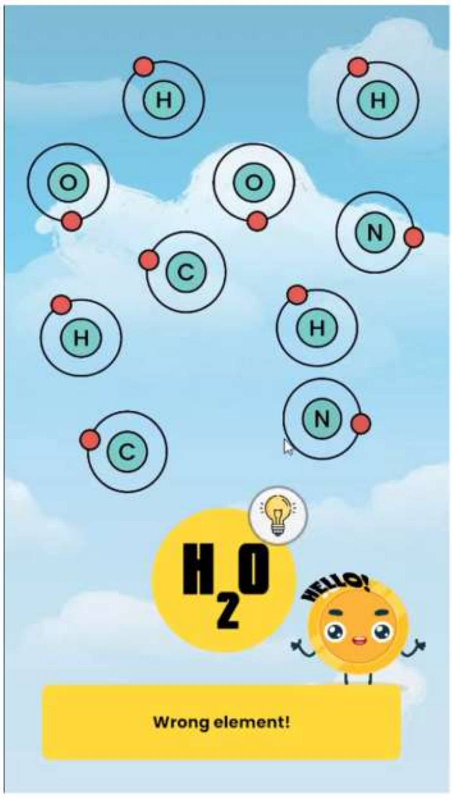
#fff

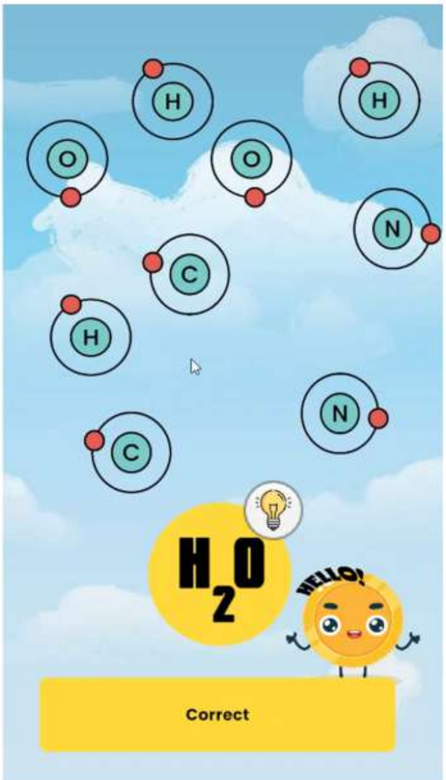
These are color values that I have use to desing the game

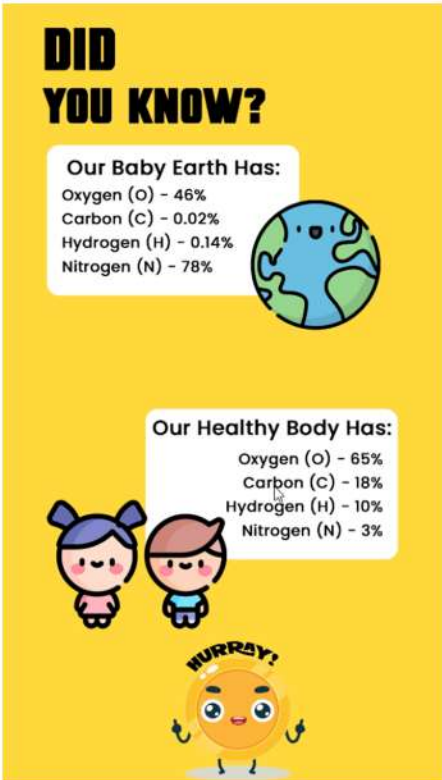
Game Output:

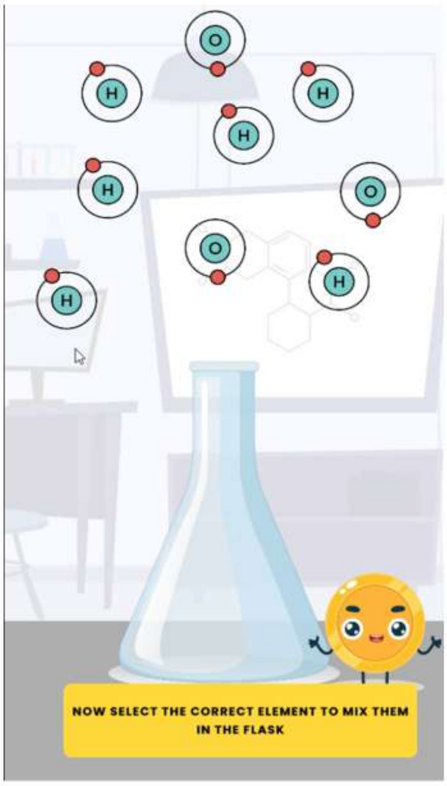

Chem
LAB
Learn Everything Here

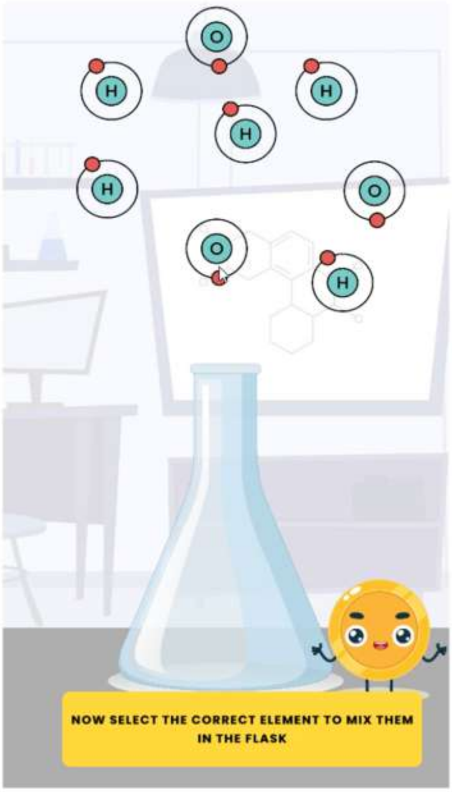

H_2O
Hai I am Mittu, your chemistry friend here
I will help you here to learn chemistry

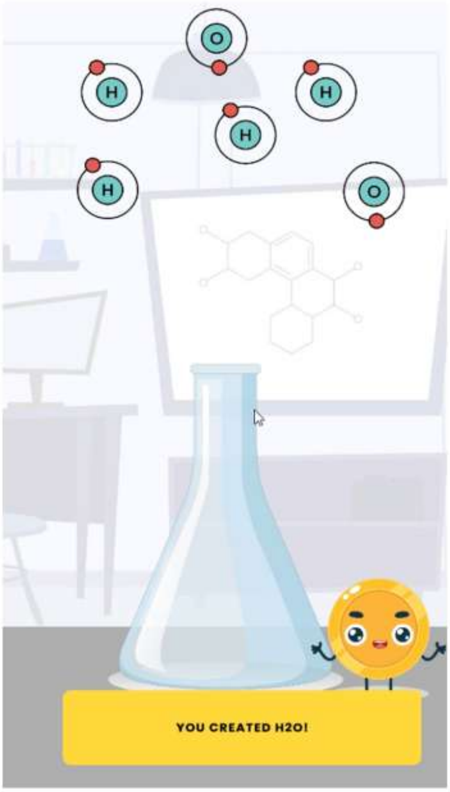

H_2O
Wrong element!

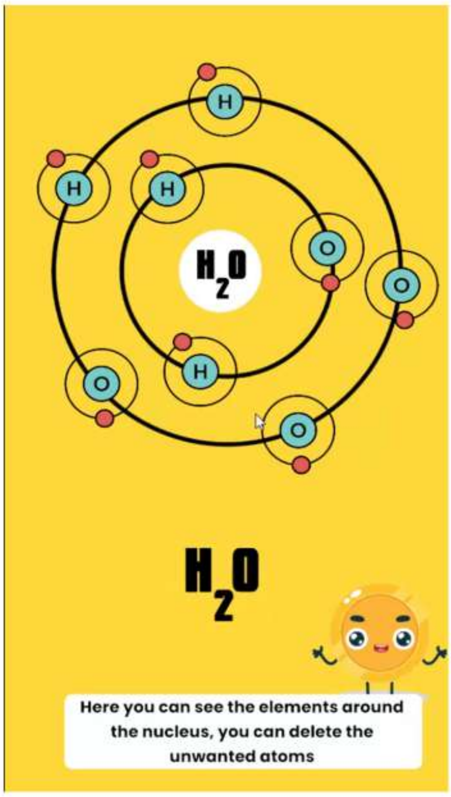

H_2O
Correct

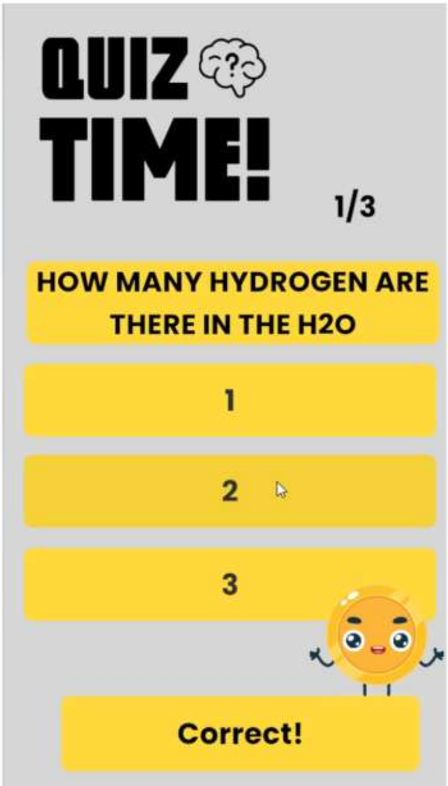

DID
YOU KNOW?
Our Baby Earth Has:
Oxygen (O) - 46%
Carbon (C) - 0.02%
Hydrogen (H) - 0.14%
Nitrogen (N) - 78%
Our Healthy Body Has:
Oxygen (O) - 65%
Carbon (C) - 18%
Hydrogen (H) - 10%
Nitrogen (N) - 3%
HURRAY!

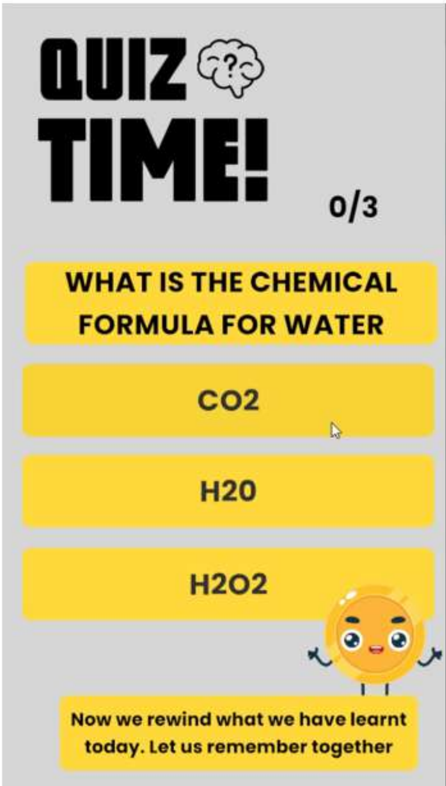

NOW SELECT THE CORRECT ELEMENT TO MIX THEM
IN THE FLASK

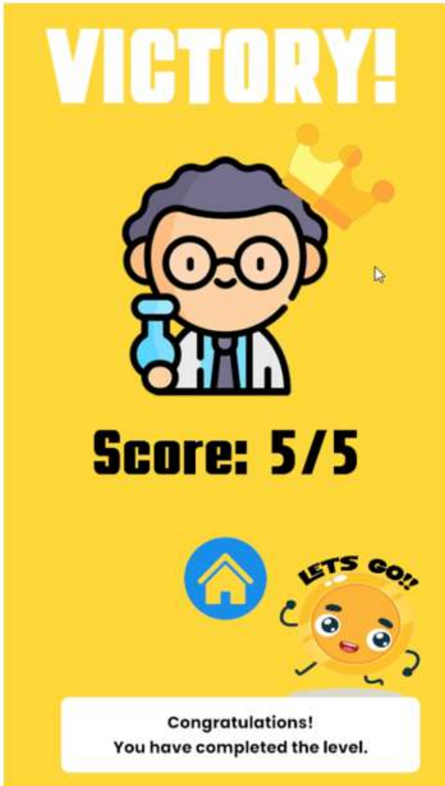

H
O
NOW SELECT THE CORRECT ELEMENT TO MIX THEM IN THE FLASK


H
O
YOU CREATED H2O!


H
O
H_2O
H_2O
Here you can see the elements around the nucleus, you can delete the unwanted atoms


QUIZ TIME!
1/3
HOW MANY HYDROGEN ARE THERE IN THE H2O
1
2
3
Correct!


QUIZ TIME!
0/3
WHAT IS THE CHEMICAL FORMULA FOR WATER
CO2
H20
H2O2
Now we rewind what we have learnt today. Let us remember together


VICTORY!
Score: 5/5
LETS GO!!
Congratulations!
You have completed the level.